



MAT1503 EXAM Revision - Exam papers

Linear Algebra (University of South Africa)



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Exam Preparation

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MAT1503: LINEAR ALGEBRA I

Exercise 1

Question 1.

Assume we are given the system

$$\begin{cases} x & +y & -z & +2t = 2 \\ 2x & & +z & -t = 3 \\ & -y & & +2t = 2. \end{cases}$$

Which of the following elements are solutions of the above system?

- (1) (1,-1,0,1)
- (2) (1,-1,0,1, 0)
- (3) (0,1,3,1)
- (4) (2,-3,3,1)

Question 2

Consider the system

$$\begin{cases} x & +2y & & = 1 \\ 3x & +ky & & = 3 \\ x & +ky & +z & = 2 \end{cases}$$

- (1) For which value(s) of k is $(1, 0, 1)$ a solution to the above system?
- (2) For which value(s) of k is $(1, 0, -1)$ a solution to the above system?
- (3) For which value(s) of k is $(3, -1, 5)$ a solution to the above system?

Additional Exercises

Same instructions as Question 2.

(1)

$$\begin{bmatrix} 3 & 2 & 1 \\ 0 & k+1 & 3 \\ 0 & 0 & k-3 \end{bmatrix}$$

(2)

$$\begin{bmatrix} \lambda & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & \lambda+3 \end{bmatrix}$$

(3)

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & k-1 & 3 \\ 0 & 0 & (k-1)(k+2) \end{bmatrix}$$

Exercise 2

Suppose A , B and C are 2×2 matrices such that $\det(A) = 3$, $\det(B^{-1}) = -2$ and $\det(C^T) = 4$.

Evaluate

- (a) $\det(ABC)$
- (b) $\det([C^2]^T)$
- (c) $\det(-4A)$
- (d) $\det(-4A^{-1})$
- (e) $\det([-4A]^{-1})$.

Exercise 3

For which values of k is the coefficient matrix of the system given in Question 2, Exercise 1 invertible?

Exercise 4

Let $A = \begin{bmatrix} -1 & 5 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ -2 & 1 \end{bmatrix}$,

- (i) Compute
 - (a) A^{-1} , B^{-1} and AB
 - (b) AA^{-1} , BB^{-1} and $A^{-1}B^{-1}$
 - (c) A^2 , B^2 and $A^2 - B^2$.
- (ii) Using (a), determine a matrix Z such that $ZA = B$
- (iii) Using (a) determine a matrix Y such that $AY = B$.

Exercise 5

(1) Solve the following system with Cramer's Rule

$$\begin{cases} 2x_1 + x_2 + x_3 = 3 \\ x_1 - x_2 - x_3 = 0 \\ x_1 + 2x_2 - x_3 = 0 \end{cases}$$

(2) Use cofactors to find the determinant of the coefficient matrix of the system.

Exercise 6

Let $\vec{a}$ and $\vec{b}$ be the vectors given by $\vec{a} = \langle 2, 0, -1 \rangle$ and $\vec{b} = \langle 1, -1, 3 \rangle$.

- (1) Compute
 - (i) $\|\vec{a}\|$ and $\|\vec{b}\|$
 - (ii) $\vec{a} \cdot \vec{a}$, $\vec{a} \cdot \vec{b}$ in two different ways.
 - (iii) $\vec{a} \times \vec{b}$ and $\vec{b} \times \vec{a}$
- (2) Assume that θ is the angle between the two vectors $\vec{a}$ and $\vec{b}$. Write down $\cos \theta$ in terms of $\vec{a} \cdot \vec{b}$ and $\|\vec{a}\|$ and $\|\vec{b}\|$. Deduce $\cos \theta$ and find the angle θ between the two vectors.
- (3) Find an equation of the plane P containing $\vec{a}$ and $\vec{b}$
- (4) Calculate the area of the parallelogram determined by $\vec{a}$ and $\vec{b}$
- (5) Determine the parametric equations of the plane containing $\vec{a}$ and $\vec{b}$
- (6) Determine the equation of the plane Q parallel to the plane P in (3) and passing through point
 - (i) $(0, 0, 1)$
 - (ii) $(0, 1, 0)$
 - (iii) $(1, 0, 1)$
 - (iv) $(1, 1, 0)$
 - (v) $(0, 1, 1)$
 - (iv) $(-1, 1, -1)$
- (7) Find an equation of the line L perpendicular to P in (3) and passing through the tip (terminal point) of the vector $\vec{u}$ where $\vec{u}$ is
 - (i) the unit vector in the direction of $\vec{a}$ with initial point $P_0 = (2, 0, -1)$
 - (ii) the unit vector in the direction of $\vec{b}$ with initial point $P_1 = (1, -1, 3)$.
- (8) Find a point-normal form of the equation of the plane passing through the point $P(1, -2, -1)$ and having normal $\vec{n} = \langle 1, -1, 4 \rangle$
- (9) Determine whether the planes $x + 2y - 7z = 3$ and $-2x + y - 9 = 0$ are perpendicular or parallel.

Exercise 7

- (1) Find all the roots (denote them by z_1 and z_2) of $(\sqrt{2} - i\sqrt{2})^{1/2}$ and sketch them as vectors in the complex plane. Write z_1 and z_2 in
 - (a) standard form
 - (b) polar form
- (2) Use your result from 1(a) to find z_1/z_2 and $\overline{z_1/z_2}$

- (3) Use de Moivre's Theorem to express $\sin(2\theta)$, $\sin(3\theta)$, $\cos(2\theta)$, $\cos(3\theta)$ and $\cos(4\theta)$ in terms of powers of
- (a) $\sin \theta$ and $\cos \theta$.
 - (b) $\sin \theta$ only
 - (c) $\cos \theta$ only
- (4) Determine the 4^{th} roots of 16 in
- (i) standard form
 - (ii) polar form
- (5) Let $z_1 = 2 - 3i$, $z_2 = -1 + 4i$. Find
- (i) $\overline{z_1}$, $\overline{z_2}$, $\overline{z_1 - z_2}$, $\overline{z_1 + z_2}$, $\operatorname{Re}(z_1)$, $\operatorname{Re}(z_2)$ and $\operatorname{Re}(z_1 z_2)$
 - (ii) $\frac{z_1}{z_2}$ and $\frac{\overline{z_1}}{\overline{z_2}}$.
 - (iii) Deduce the principal arguments of $\frac{z_1}{z_2}$ and $\frac{\overline{z_1}}{\overline{z_2}}$.
- (6) Use de Moivre's Theorem to determine i^3 , $(1 + i)^{100}$ in standard and polar form.

Exercise 8

Consider the points $P_0(-1, 2)$, $P_1(1, 3)$, $P_2(3, -2)$ and $P_3(1, 1)$.

Assume that the points P_i , $i = 0, 1, 2, 3$ have position vectors r_i , $i = 0, 1, 2, 3$.

- (i) Find the points $-P_0 - P_1 + P_2$
- (ii) Solve the system

$$xr_0 + yr_1 + zr_2 = r_3.$$

- (iii) Find z whenever $x = y$.

Exercise 9

- (a) Let

$$A = \begin{bmatrix} 7 & 2 & 1 \\ 3 & 4 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } b = \begin{bmatrix} z - 2 \\ 3x - z + 1 \end{bmatrix}. \text{ Verify that } A = -A + 2A.$$

Find x , y and z such that $AX = b$.

Exercise 10

Write your own Summary or Mind Map for each of the following tables

①

Linear systems of equations

②

Matrices

③

finding the inverse A^{-1} of a square matrix A

Elementary Matrices

④

Determinant

⑤

Invertibility of Matrices